

1(a)(i)	hazard <u>and</u> risk <u>and</u> precaution ;	1												
	<table> <tr> <th>hazard</th><th>risk</th><th>precaution</th></tr> <tr> <td>chemical / anaesthetic</td><td>toxic / allergy / irritant</td><td>use, gloves / goggles / mask / PPE / use fume cupboard</td></tr> <tr> <td><i>Drosophila</i> / fruit flies</td><td>allergy / (spread of) pathogens / biohazard</td><td>use gloves / goggles / mask / PPE</td></tr> <tr> <td>agar jelly / specimen tube, containing food</td><td>allergy / biohazard / pathogens</td><td>use gloves / goggles / mask / PPE / aseptic technique</td></tr> </table>	hazard	risk	precaution	chemical / anaesthetic	toxic / allergy / irritant	use, gloves / goggles / mask / PPE / use fume cupboard	<i>Drosophila</i> / fruit flies	allergy / (spread of) pathogens / biohazard	use gloves / goggles / mask / PPE	agar jelly / specimen tube, containing food	allergy / biohazard / pathogens	use gloves / goggles / mask / PPE / aseptic technique	
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1(a)(ii)	any seven from: 1 (to produce heterozygotes) (breed) brown-eyed / AAdd, females with, scarlet-eyed / aaDD, males or (breed) scarlet-eyed / aaDD, females with, brown-eyed / AAdd, males ; 2 <u>method</u> to maintain a, constant / stated, temperature ; 3 <i>idea of</i> remove parent (fruit flies) from their offspring, before they are adults / before day 10 / before eggs hatch ; 4 put (anaesthetised), double heterozygotes / offspring from first cross, into (new specimen) tube ; 5 (cross double heterozygotes to give) a large number of <u>offspring</u> ; 6 use, stated / same / recommended / correct, dose / volume / concentration, (of chemical / anaesthetic) ; 7 identify, phenotype / eye colour, using, microscope / magnifying glass / hand lens ; 8 count / record, the number of flies of each, phenotype / eye colour ; 9 <i>idea of</i> method to prevent double counting (of flies) ;	7												

Question	Answer	Marks
1(a)(iii)	9 : 3 : 3 : 1 ; red : brown : scarlet : white ;	2
1(b)	any three from: 1 critical value (at degrees of freedom = 3 and $p = 0.05$) is 7.815 ; 2 calculated (χ^2) value / 4.798, is less than, critical value / 7.815 ; 3 null hypothesis is accepted (at $p = 0.05$) ; 4 there is <u>no significant difference</u> (at $p = 0.05$) ;	3